Build a template with building blocks

Practical clinical modelling with the Nursing Admission Assessment

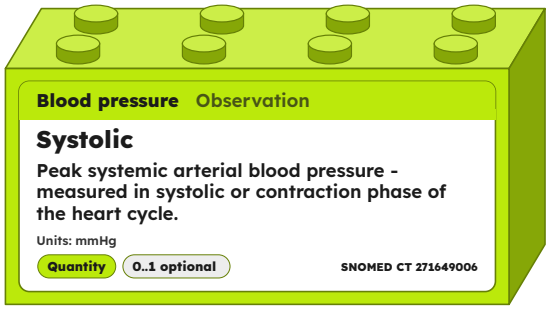
In this practical you build the **Nursing Admission Assessment** template on paper, block by block. Archetypes are deliberately broad so they can be reused everywhere; template design means narrowing each archetype down to what this specific form needs, and pulling in extra archetypes for what is still missing.

Based on the openEHR practical modelling tasks by Ian McNicoll (freshEHR), with additions by the Hanze for EHRCON26. If you want to explore further, the online version of the workshop is at **freshehrteam.github.io/openehr-workshop**

Your materials

Building blocks

- **One building block = one data element** of an archetype. Same colour = same archetype.
- **Header:** the archetype it belongs to, and its entry type.
- **Chips:** the data type, and the cardinality the archetype allows: **0..1** optional, **1..1** required, **0..*** repeats.
- **Grey text** under the name: what the element is for, taken from the archetype.
- **Code at the bottom:** a SNOMED CT or LOINC code from the archetype, if it has one.



The baseplate, headings and plates

- **Green Encounter baseplate = the template** (COMPOSITION). Lay it along the edge of your table: you build the template up from it, seen from the front, like a wall of building blocks.
- **Grey plates = Ad hoc heading** (SECTION): a generic heading; write its name on the plate.
- **Coloured plates = archetypes.** One per archetype, in its own colour, two building blocks wide. Write the cardinality the archetype gets in your template on the plate; a new name goes on a yellow sticker over the archetype's name.
- **Everything stands on what it belongs to**, lined up with the studs: building blocks on their archetype plate, an archetype plate on a heading or on the baseplate. The higher up, the deeper in the template.

Changing the template

- **Remove an element** from the template (constrain it out, 0..0): take its building block off the template. It stays in the archetype, just not in this form.
- **Yellow stickers = template changes:** a new name, or a stricter cardinality. A template can only **narrow** what the archetype allows: going from 0..1 to 1..1 is fine, going from 0..1 to 0..* is not.

Exercise 0: Choose the archetypes

Start with the big picture: which archetypes does this form need? Use only the green Encounter, the grey headings and the archetype plates for now, and keep the building blocks aside.

- Build on the green **Encounter** baseplate: the parts of the form from left to right, in the same order as on the form, side by side with one free stud between them.

- Not every part of the form needs a grey heading: an archetype's name already works as a heading (and you can rename it to match the form). Use a heading (**SECTION**) only to group several archetypes, or to put one heading above a list that repeats. Write the names on the grey plates.
- Several archetypes under one heading? Stack them on the heading, one on top of the other.
- Find an archetype plate for each part of the form. Not every plate belongs in your template, and some parts need more than one.
- Read the purpose and the entry type (OBSERVATION, EVALUATION, INSTRUCTION, CLUSTER) on each plate: does it fit what the form records?
- Can the form hold more than one of something? Fill in the cardinality on that archetype's plate.
- Found a dashed CLUSTER plate you need? Put it on top of the archetype it slots into.
- Is there something on the form that none of the plates covers? Go to the **CKM desk** (your facilitator) and describe what you're looking for, as precisely as you can.
- Medications: in openEHR, the list of medication a patient uses is made of medication orders, even when someone else (such as the GP) prescribed them.

Exercise 1: Choose the elements

Now take the building blocks of the archetypes you chose, and put them on their plates. Archetypes are deliberately broad, so most blocks won't be needed: narrow each archetype down to what this form needs. We do 1.1 together; from 1.2 on, the steps say what the form needs, not which block to use.

1.1 Main diagnosis (together)

- Rename the **Problem/Diagnosis** archetype to **Main Diagnosis**: write the new name on a blank yellow sticker and stick it over the name on its plate.
- Constrain out everything apart from **Problem/Diagnosis name**: take the other blocks off the plate.

1.2 Patient history, medication and vital signs

- Keep only the blocks the form needs for **Patient history**, the **name of the medication**, and the vital signs **Height**, **Weight** and **Blood pressure**.
- Which blood pressure values should never be left empty? Make them mandatory.

1.3 Allergies

- Keep only the element that says what the patient is allergic to, and rename it to match the form.
- The reaction is a CLUSTER. Find the building block on the allergy archetype that it slots into (the CLUSTER slot: its studs have the colour of the reaction plate), and put the reaction plate on top of that block. The plate is wider than the block: let it hang over the side where there is room.
- Keep only the element that describes how the reaction showed itself. Rename it to match the form, and make it mandatory.

1.4 Medication directions

In the **medication** list on the form, the 'Directions' column holds the prescriber's directions for each medication: how much, and how often. There is no element called Dose or Frequency.

- Look at the **Medication order** elements about directions and read their descriptions. Which one covers the whole of the directions?
- Keep that element and rename it to match the form.

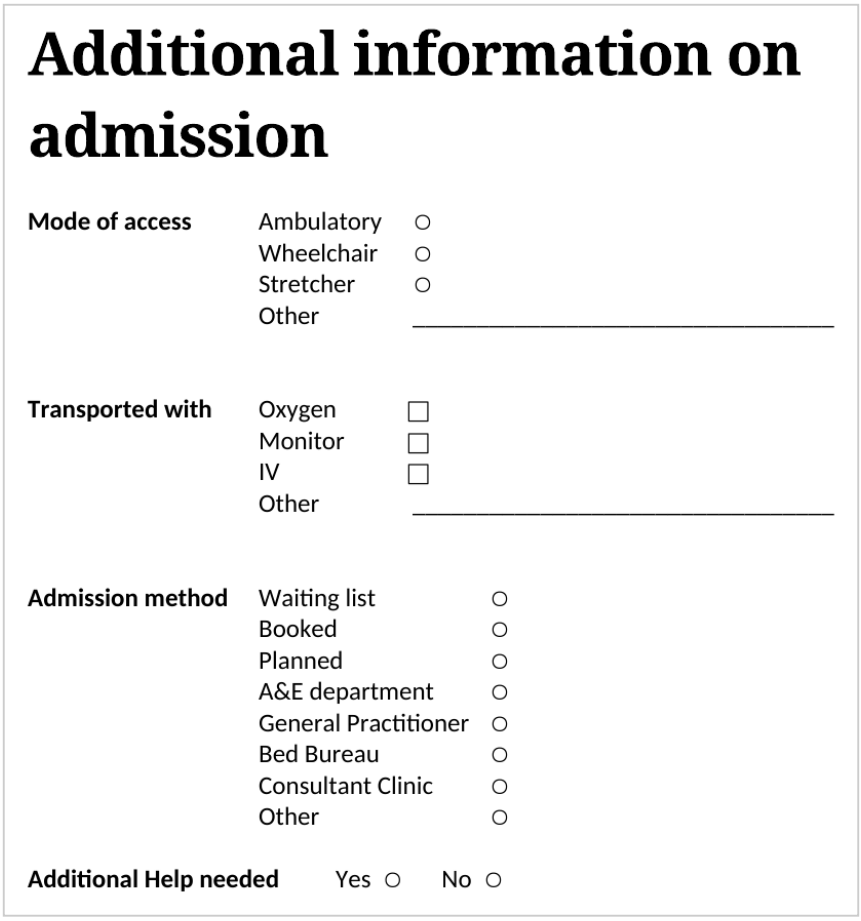
1.5 Frailty

- Keep the element that holds the frailty score.

Exercise 2: Create a new local archetype

The nurses want to record **additional information on admission** (paper form below). CKM has no suitable archetype, so you make a new **ADMIN_ENTRY**: admission logistics, not a clinical finding.

- Get the black plate and the **write-on building blocks** from the CKM desk: archetype **Inpatient admission details**.
- Make one building block per question on the form: write its name and values, tick the data type, and fill in the cardinality.
- Which data type lets a nurse pick from a list, or write something under 'Other'?
- Can more than one 'Transported with' apply? What does that mean for the cardinality?
- Where would your new archetype go in the template: under a heading, or straight on the Encounter? It doesn't need to go on the baseplate.



Additional information on admission

Mode of access
Ambulatory ○
Wheelchair ○
Stretcher ○
Other _____

Transported with
Oxygen ☐
Monitor ☐
IV ☐
Other _____

Admission method
Waiting list ○
Booked ○
Planned ○
A&E department ○
General Practitioner ○
Bed Bureau ○
Consultant Clinic ○
Other ○

Additional Help needed Yes ○ No ○

Paper form: Additional information on admission

The paper form: Nursing Admission Assessment

Nursing Admission Assessment

Main Diagnosis _____

Patient History	
Patient's story: (in their own words)	

Allergies					
Allergy (substance)	Reaction (symptoms)	Allergy (substance)	Reaction (symptoms)	Allergy (substance)	Reaction (symptoms)

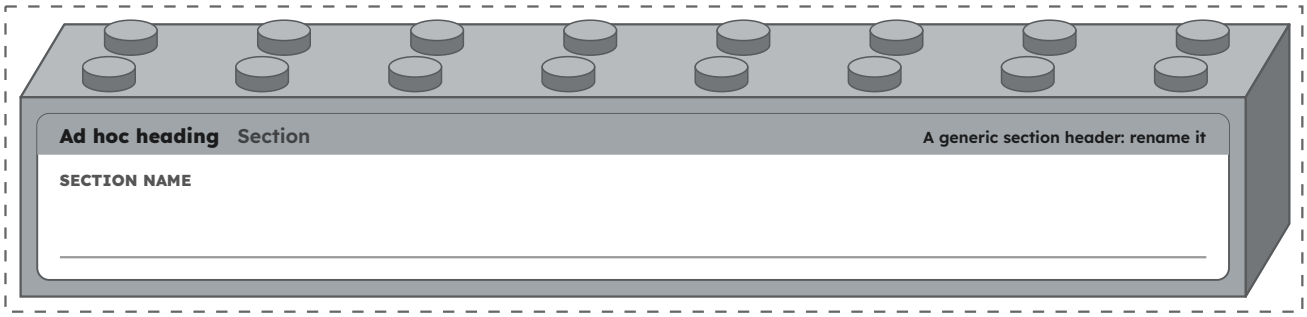
Medications			
Name	Directions (dose/frequency)	Name	Directions (dose/frequency)

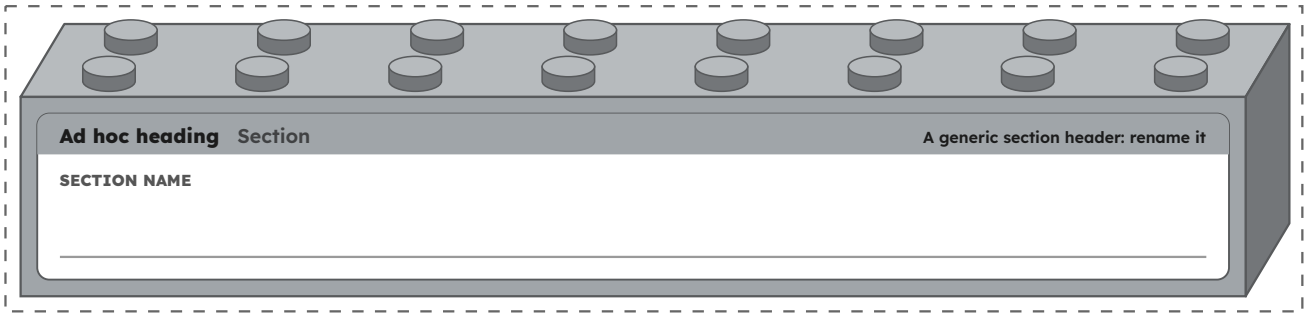
Vital Signs					
Height	cm	Weight	kg	Blood Pressure	/

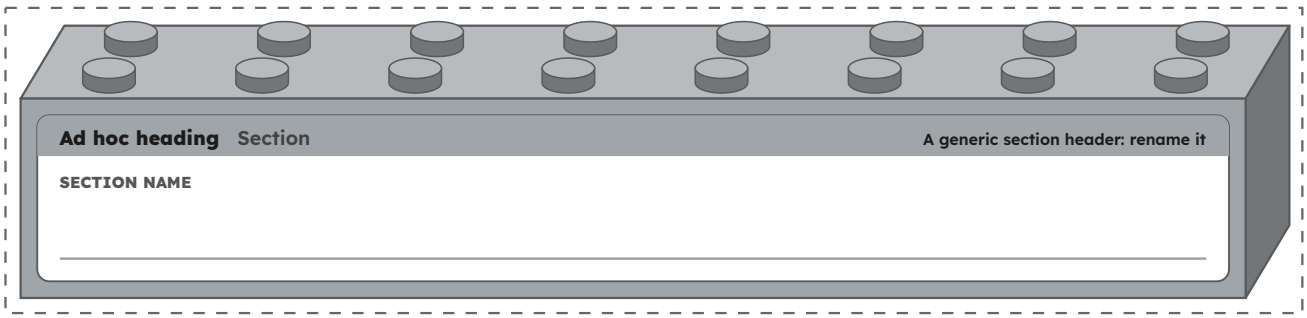
Clinical Frailty Scale (score 1–9)

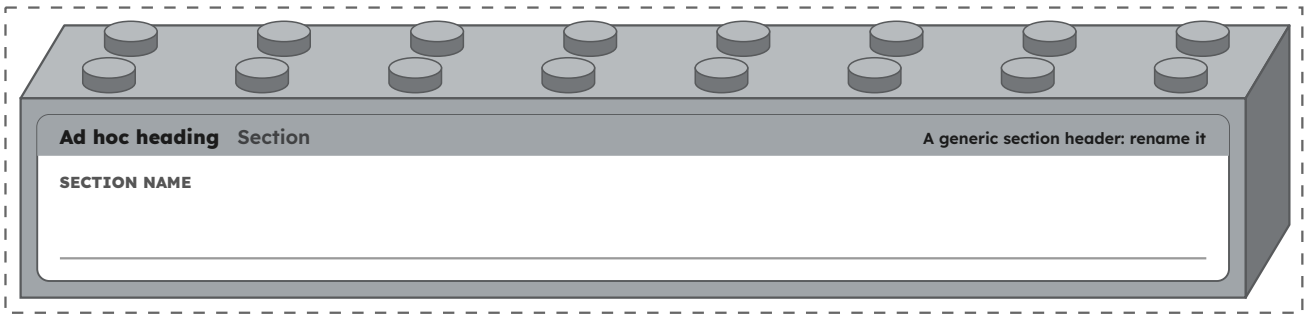
Template structure: Ad hoc headings

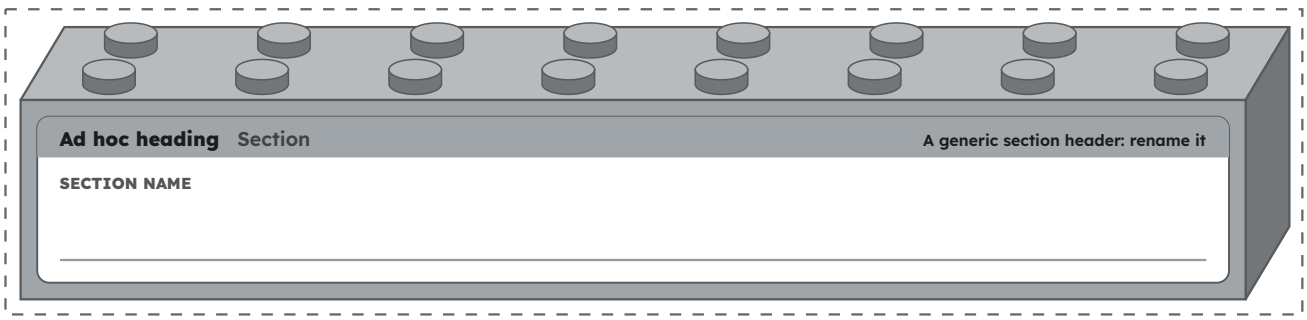
✂ cut along the dashed lines











Archetype plates

✂ cut along the dashed lines

Adverse reaction event

Cluster

CLUSTER: GOES IN A SLOT

Information about a specific adverse reaction event caused by exposure to a specific substance.

Plugs into: Reaction event summary (a building block of Adverse reaction risk)



8 of 19 elements shown

CARDINALITY

Adverse reaction risk

Evaluation

ARCHETYPE

Clinical assessment of the propensity for an individual to experience a harmful or undesirable physiological response if exposed, or re-exposed, to a substance.

8 of 12 elements shown

CARDINALITY

Blood pressure

Observation

ARCHETYPE

The local measurement of arterial blood pressure which is a surrogate for arterial pressure in the systemic circulation.

8 of 20 elements shown

SNOMED CT 364090009

CARDINALITY

Body Mass Index (BMI)

Observation

ARCHETYPE

Calculated measurement which compares a person's weight and height.

CARDINALITY

Body temperature

Observation

ARCHETYPE

A measurement of the body temperature, which is a surrogate for the core body temperature of the individual.

8 of 11 elements shown

CARDINALITY

Body weight

Observation

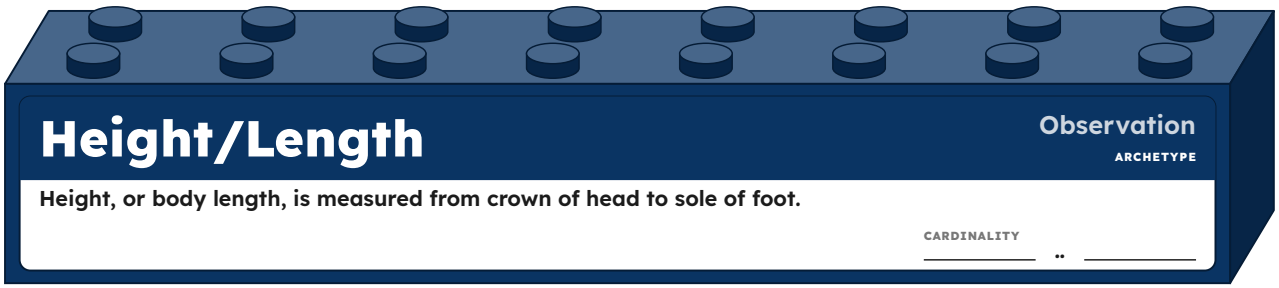
ARCHETYPE

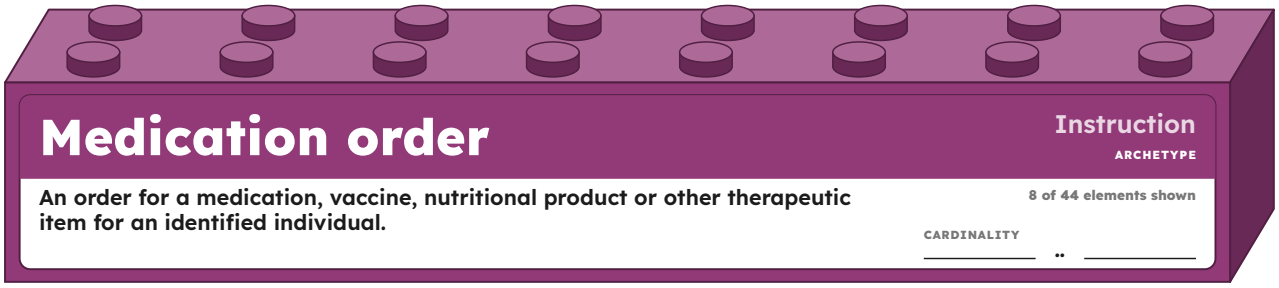
Measurement of the body weight of an individual.

CARDINALITY

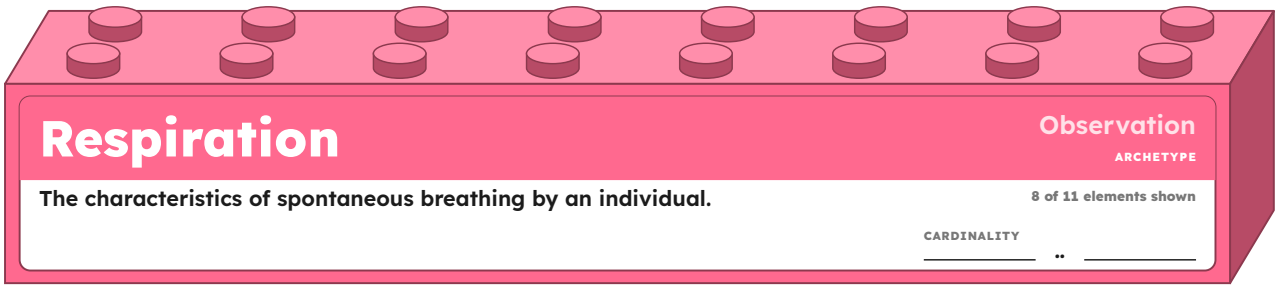
Archetype plates

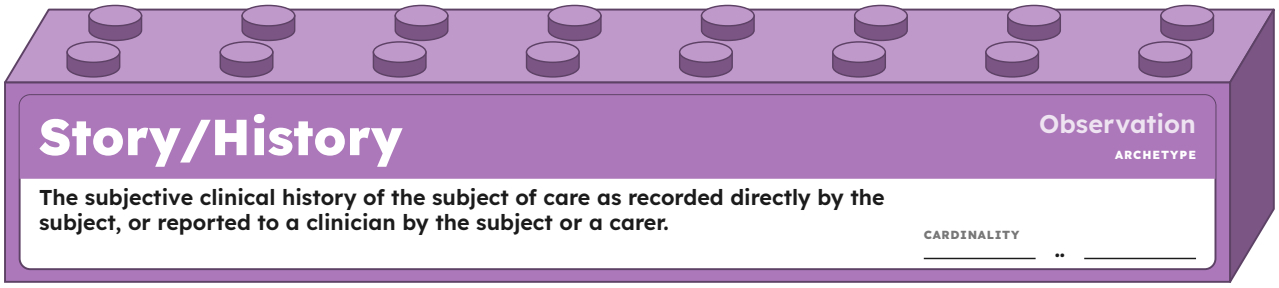
✂ cut along the dashed lines

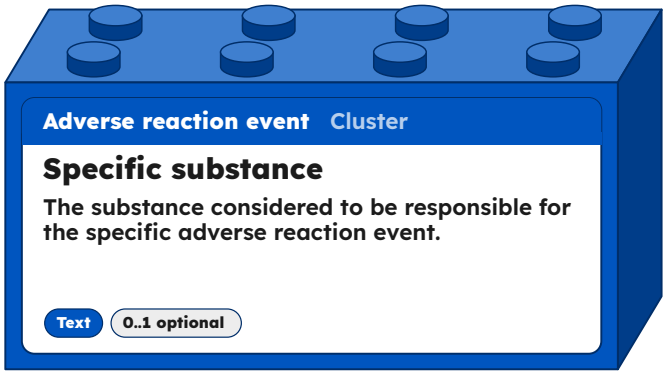


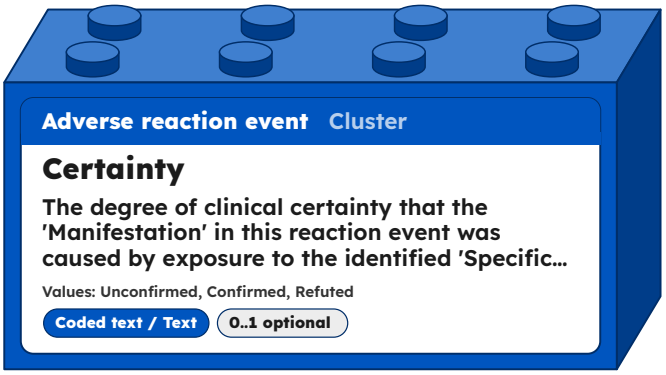


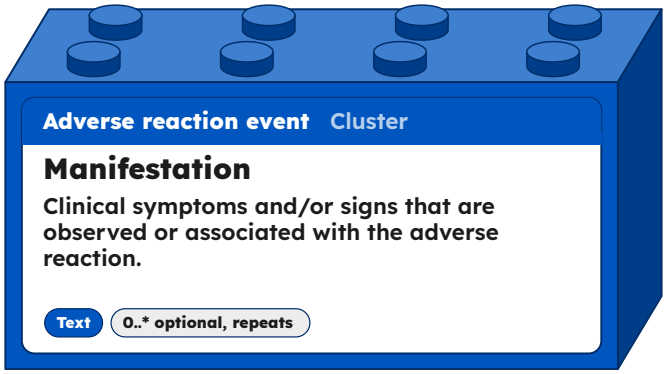


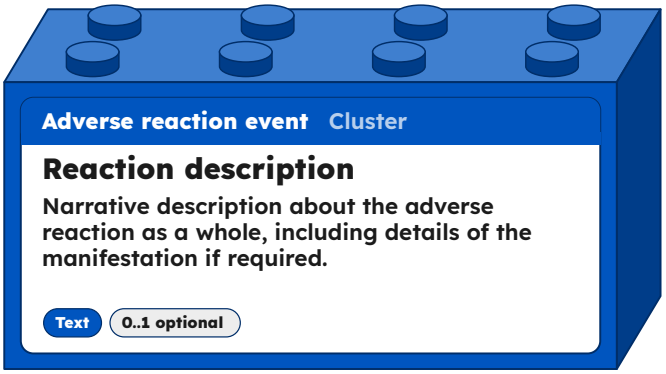


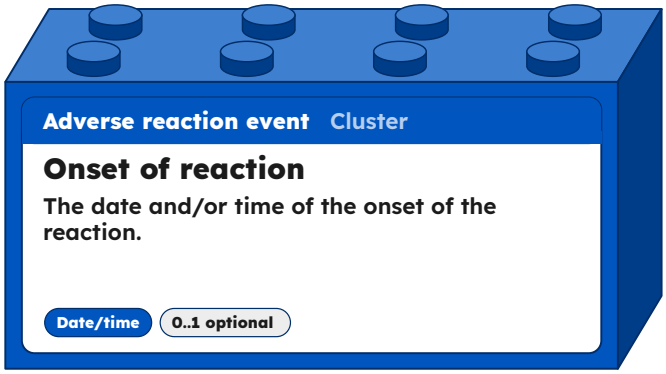


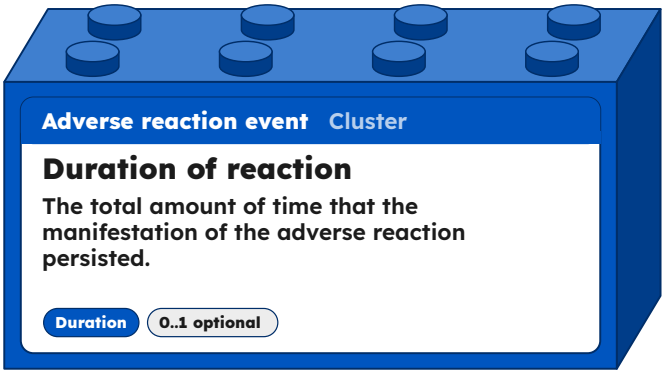


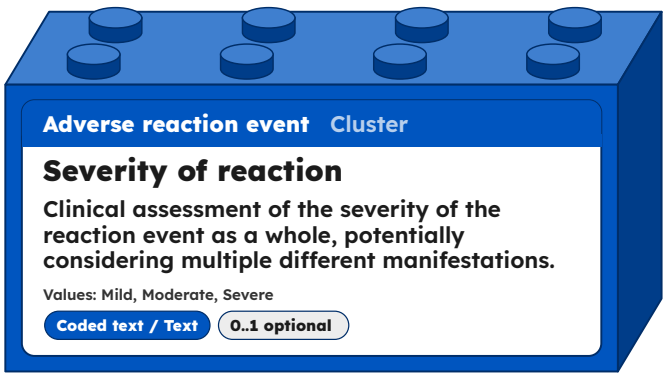


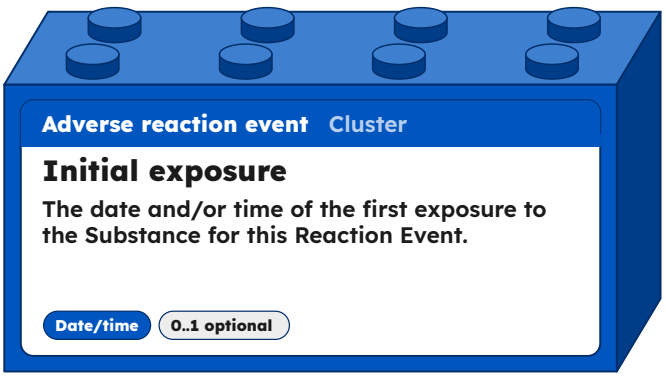












Adverse reaction event Cluster 8 of 19 elements

Adverse reaction risk Evaluation

Substance

Identification of a substance, or substance class, that is considered to put the individual at risk of an adverse reaction event.

Text | 1..1 required

Adverse reaction risk Evaluation

Active/inactive status

Status about whether the adverse reaction risk statement is active or inactive.

Values: Active, Inactive

Coded text / Text | 0..1 optional

Adverse reaction risk Evaluation

Verification status

Assertion about the certainty of the propensity, or potential future risk, of the identified 'Substance' to cause a reaction.

Values: Unconfirmed, Confirmed, Refuted

Coded text / Text | 0..1 optional

Adverse reaction risk Evaluation

Criticality

An indication of the potential for critical system organ damage or life threatening consequence.

Values: Low, High, Indeterminate

Coded text / Text | 0..1 optional

Adverse reaction risk Evaluation

Onset of last reaction

The date and/or time of the onset of the last known occurrence of a reaction event.

Date/time / Duration / Interval / Text | 0..1 optional

Adverse reaction risk Evaluation

Onset of first reaction

The onset of the first known occurrence of a reaction event.

Date/time / Duration / Text / Interval | 0..1 optional

Adverse reaction risk Evaluation

Reaction mechanism

Identification of the underlying physiological mechanism for the adverse reaction.

Values: Immune mediated, Non-immune mediated, Indeterminate

Coded text / Text | 0..1 optional

Adverse reaction risk Evaluation

Reaction event summary

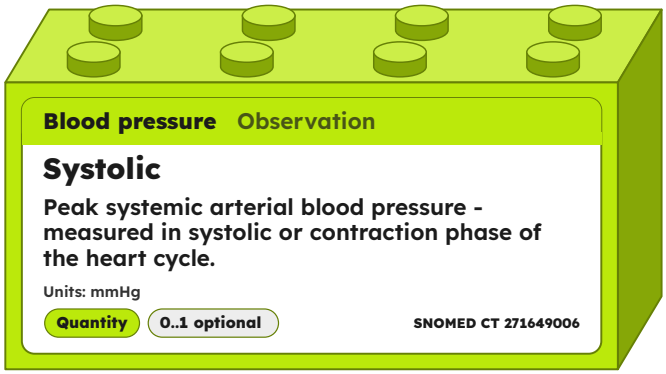
Summary details about each adverse reaction event linked to exposure to the identified 'Substance'.

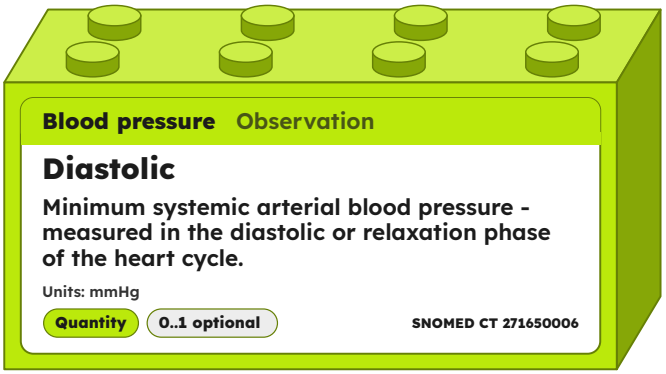


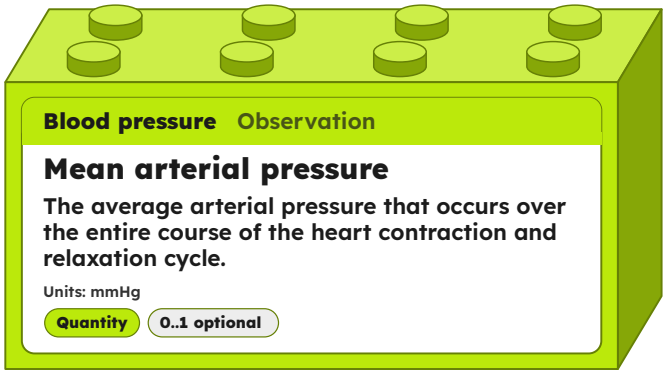
Plug in: adverse reaction event

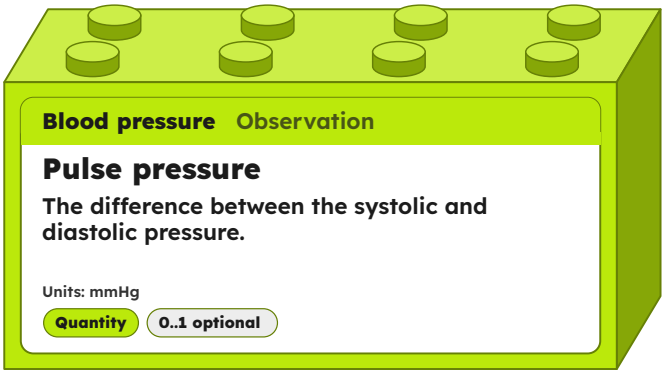
CLUSTER slot | 0..* optional, repeats

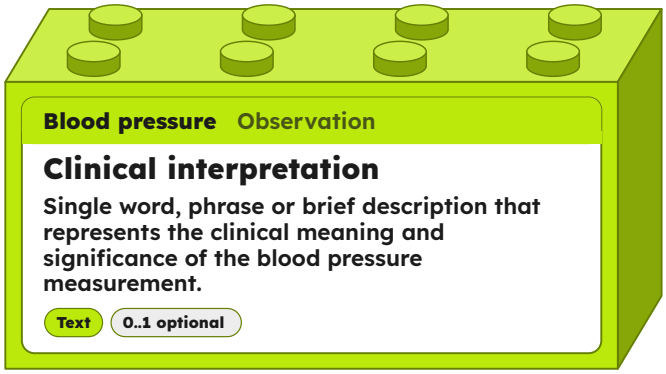
Adverse reaction risk Evaluation 8 of 12 elements

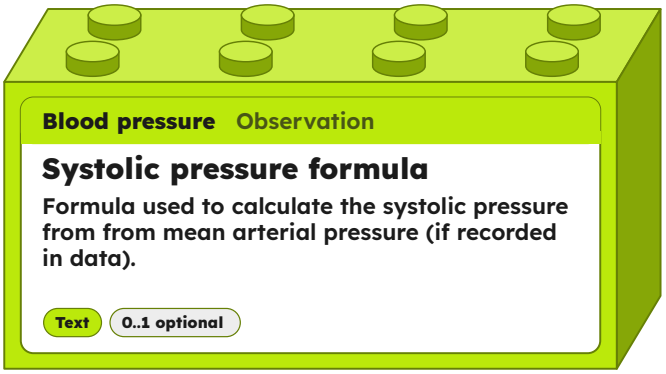

Blood pressure Observation
Systolic
Peak systemic arterial blood pressure - measured in systolic or contraction phase of the heart cycle.
Units: mmHg
Quantity
0..1 optional
SNOMED CT 271649006

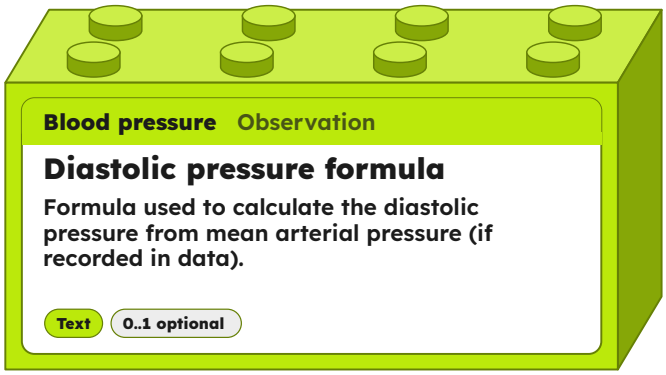

Blood pressure Observation
Diastolic
Minimum systemic arterial blood pressure - measured in the diastolic or relaxation phase of the heart cycle.
Units: mmHg
Quantity
0..1 optional
SNOMED CT 271650006

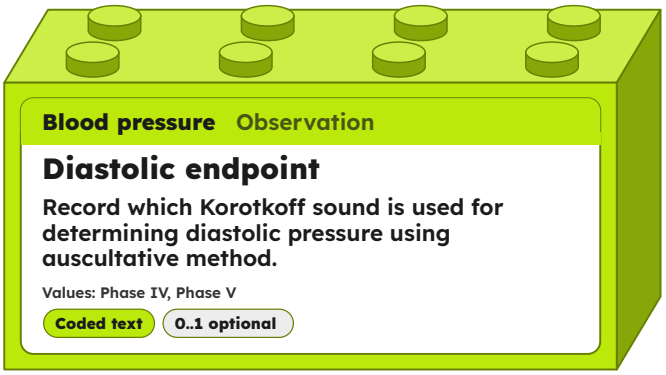

Blood pressure Observation
Mean arterial pressure
The average arterial pressure that occurs over the entire course of the heart contraction and relaxation cycle.
Units: mmHg
Quantity
0..1 optional

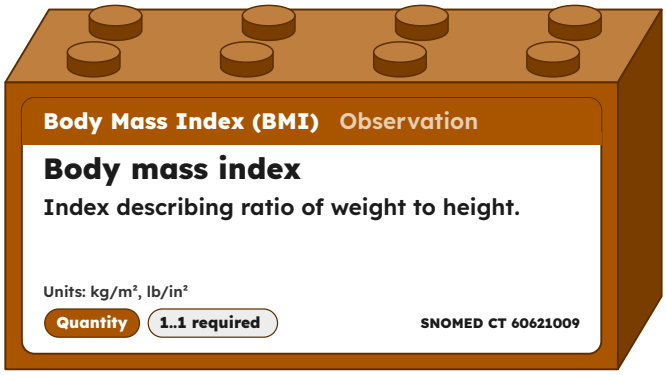

Blood pressure Observation
Pulse pressure
The difference between the systolic and diastolic pressure.
Units: mmHg
Quantity
0..1 optional

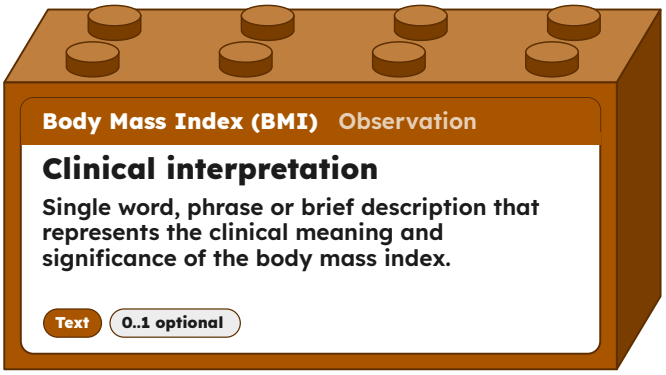

Blood pressure Observation
Clinical interpretation
Single word, phrase or brief description that represents the clinical meaning and significance of the blood pressure measurement.
Text
0..1 optional

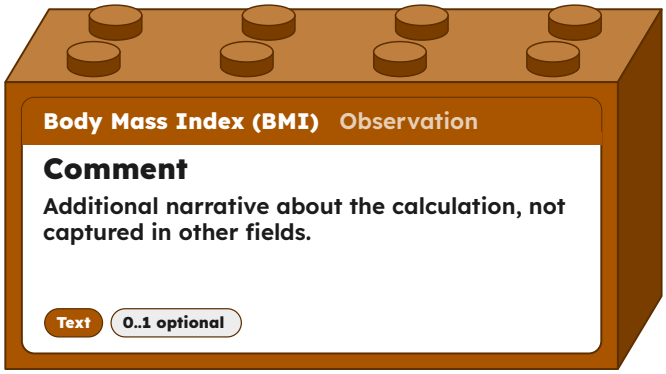

Blood pressure Observation
Systolic pressure formula
Formula used to calculate the systolic pressure from from mean arterial pressure (if recorded in data).
Text
0..1 optional

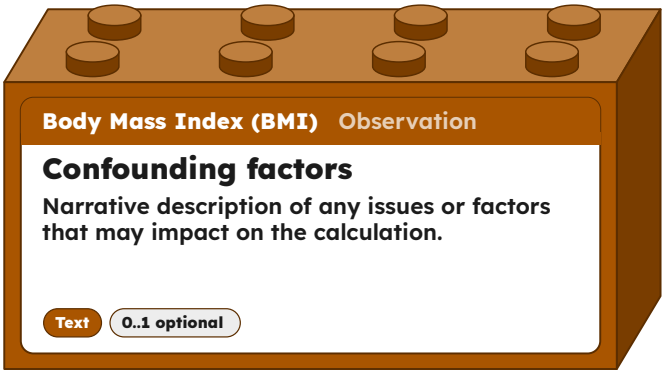

Blood pressure Observation
Diastolic pressure formula
Formula used to calculate the diastolic pressure from mean arterial pressure (if recorded in data).
Text
0..1 optional

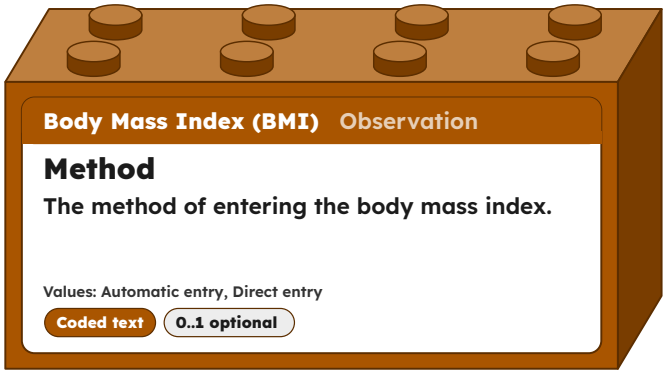

Blood pressure Observation
Diastolic endpoint
Record which Korotkoff sound is used for determining diastolic pressure using auscultative method.
Values: Phase IV, Phase V
Coded text
0..1 optional

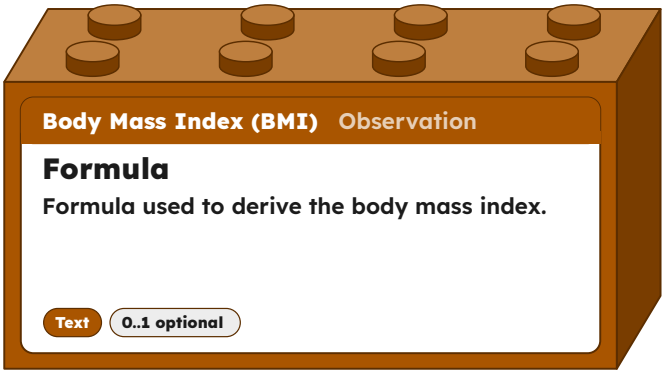

Body Mass Index (BMI) Observation
Body mass index
Index describing ratio of weight to height.
Units: kg/m², lb/in²
Quantity
1..1 required
SNOMED CT 60621009

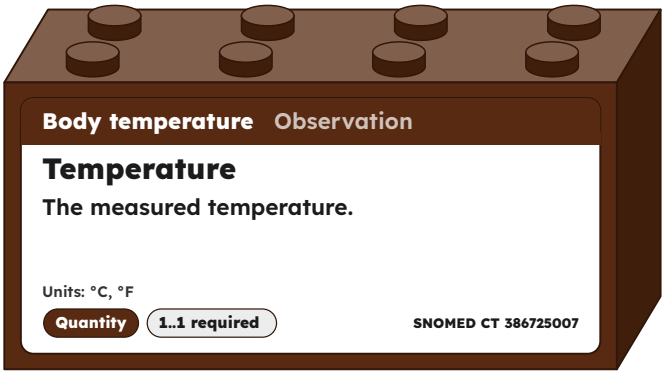

Body Mass Index (BMI) Observation
Clinical interpretation
Single word, phrase or brief description that represents the clinical meaning and significance of the body mass index.
Text
0..1 optional

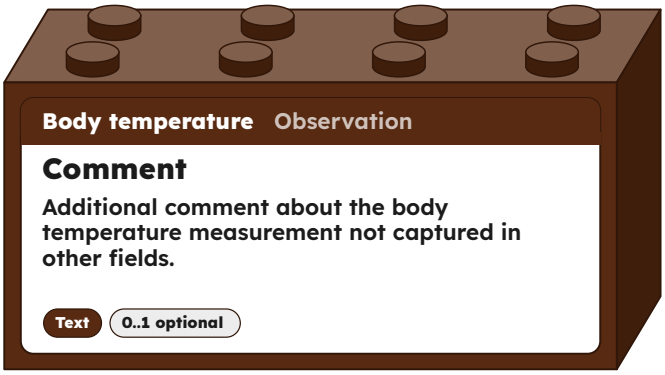

Body Mass Index (BMI) Observation
Comment
Additional narrative about the calculation, not captured in other fields.
Text
0..1 optional

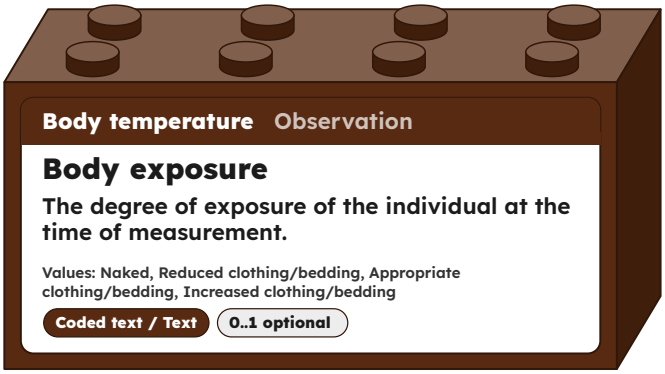

Body Mass Index (BMI) Observation
Confounding factors
Narrative description of any issues or factors that may impact on the calculation.
Text
0..1 optional

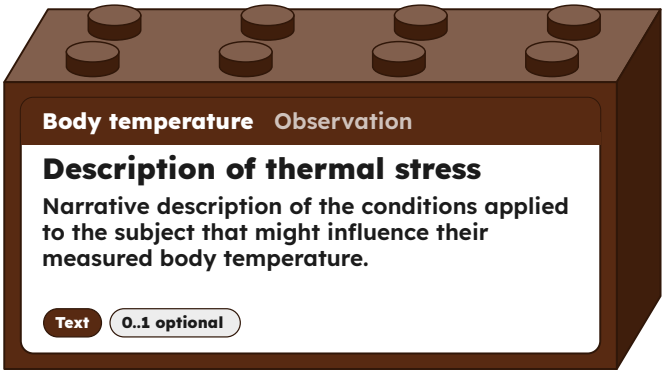

Body Mass Index (BMI) Observation
Method
The method of entering the body mass index.
Values: Automatic entry, Direct entry
Coded text
0..1 optional


Body Mass Index (BMI) Observation
Formula
Formula used to derive the body mass index.
Text
0..1 optional

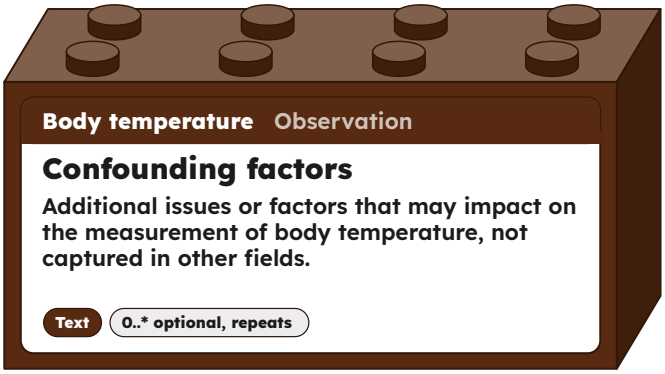


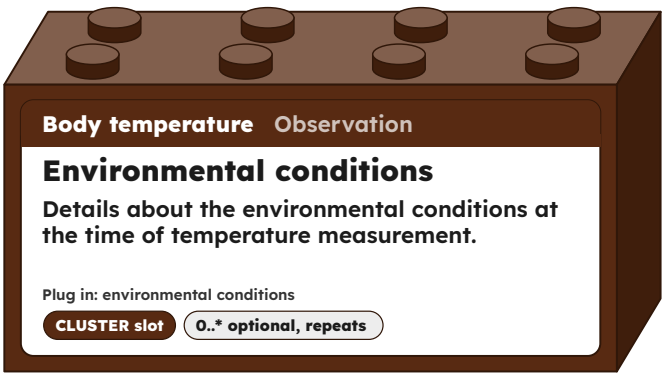


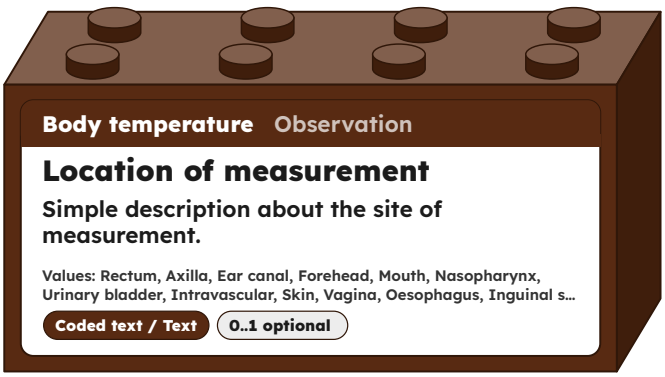




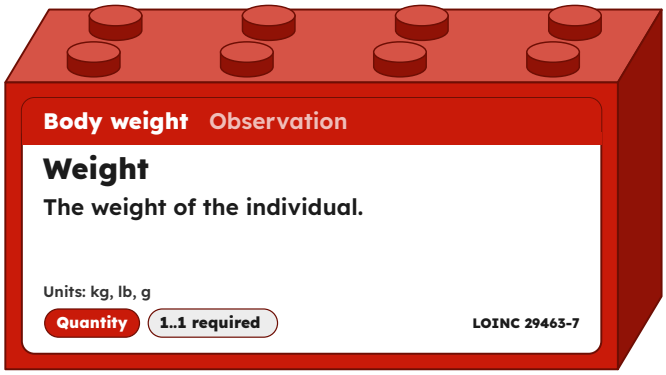


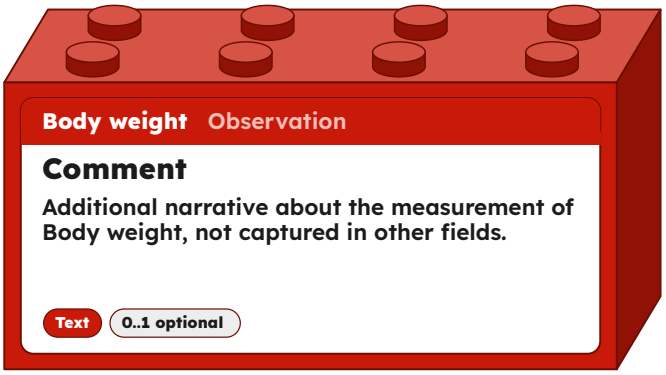




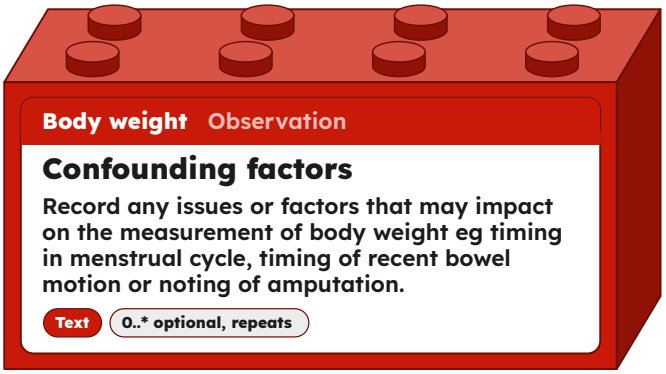


Body temperature Observation 8 of 11 elements

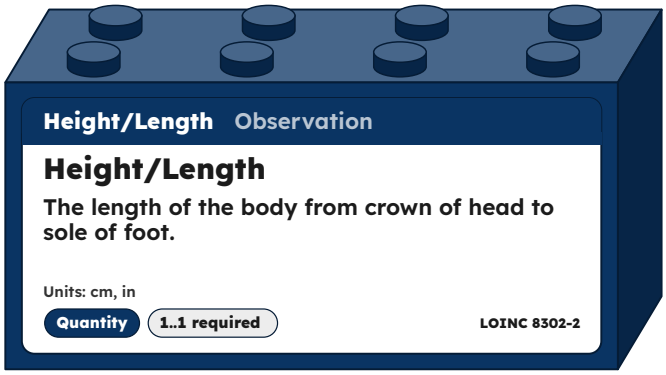

Body weight Observation
Weight
The weight of the individual.
Units: kg, lb, g
Quantity
1..1 required
LOINC 29463-7

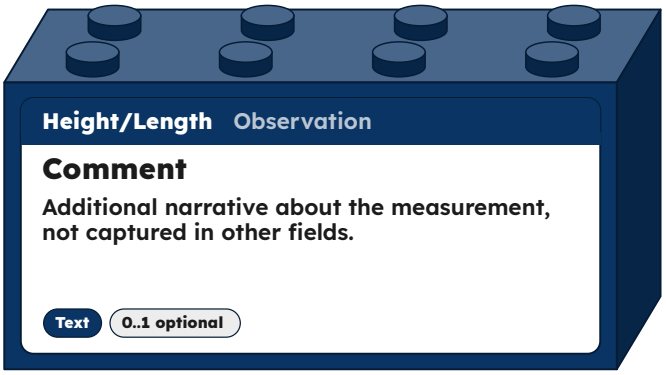

Body weight Observation
Comment
Additional narrative about the measurement of Body weight, not captured in other fields.
Text
0..1 optional

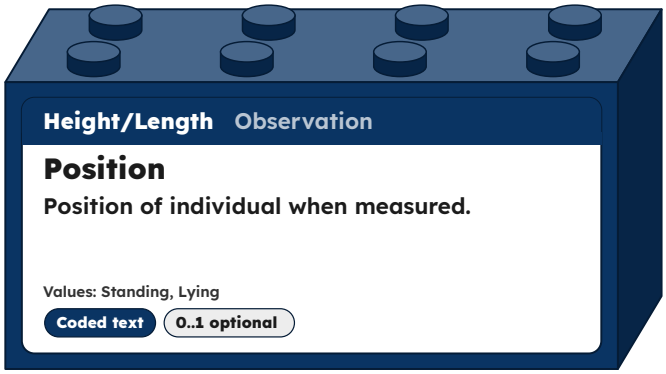

Body weight Observation
State of dress
Description of the state of dress of the person at the time of weighing.
Values: Naked; Nappy/diaper; Lightly clothed/underwear; Fully clothed, without shoes; Fully clothed, including shoes
Coded text
0..1 optional

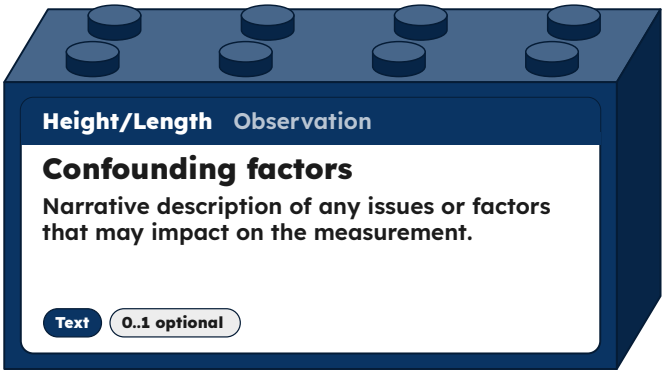

Body weight Observation
Confounding factors
Record any issues or factors that may impact on the measurement of body weight eg timing in menstrual cycle, timing of recent bowel motion or noting of amputation.
Text
0..* optional, repeats

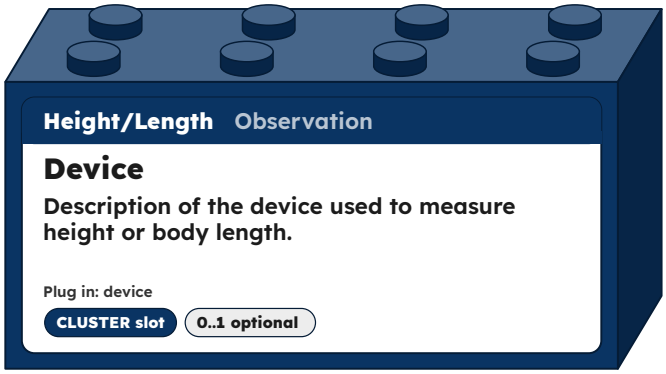

Body weight Observation
Device
Details about the weighing device.
Plug in: device
CLUSTER slot
0..1 optional

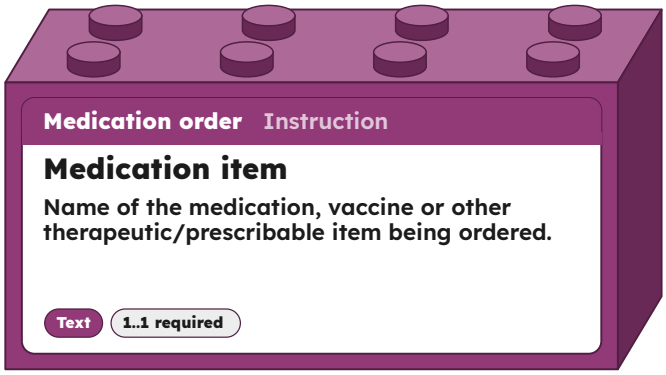

Height/Length Observation
Height/Length
The length of the body from crown of head to sole of foot.
Units: cm, in
Quantity
1..1 required
LOINC 8302-2

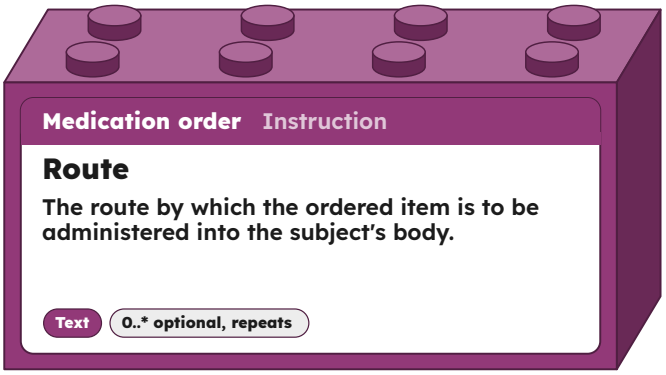

Height/Length Observation
Comment
Additional narrative about the measurement, not captured in other fields.
Text
0..1 optional

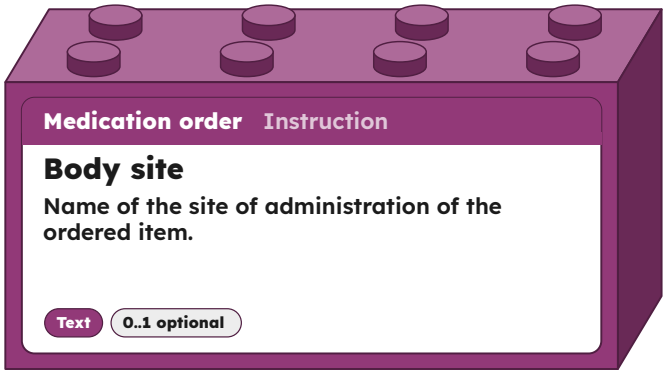

Height/Length Observation
Position
Position of individual when measured.
Values: Standing, Lying
Coded text
0..1 optional

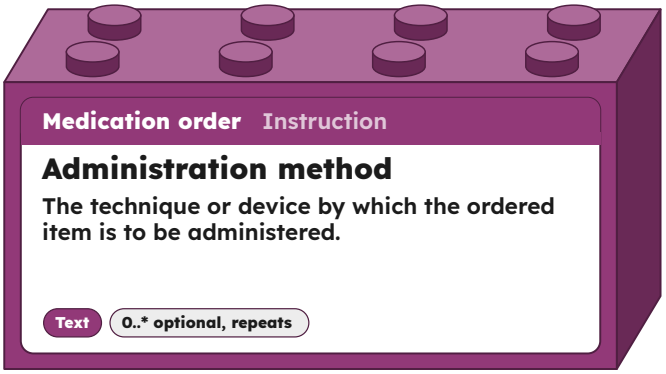

Height/Length Observation
Confounding factors
Narrative description of any issues or factors that may impact on the measurement.
Text
0..1 optional

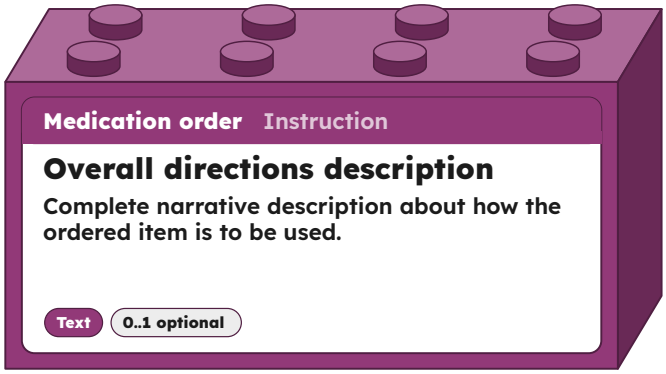

Height/Length Observation
Device
Description of the device used to measure height or body length.
Plug in: device
CLUSTER slot
0..1 optional

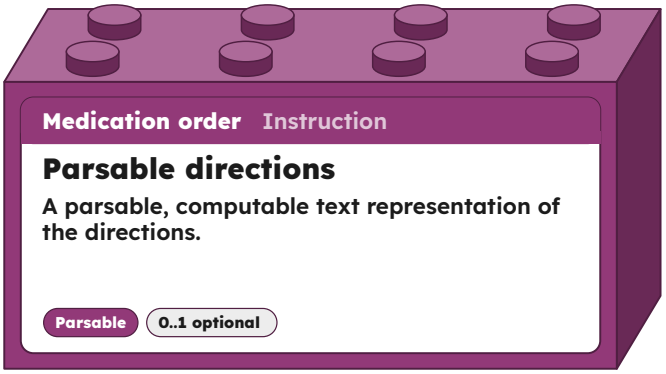


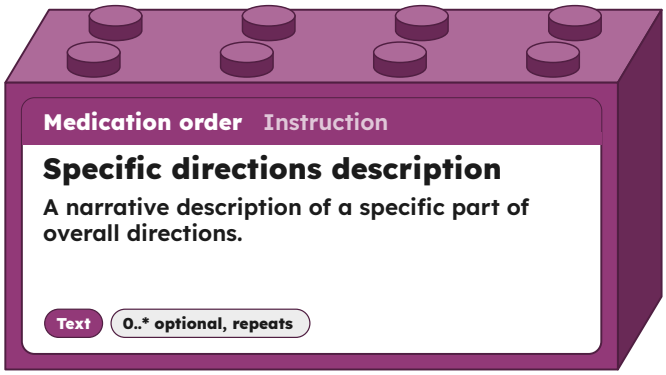


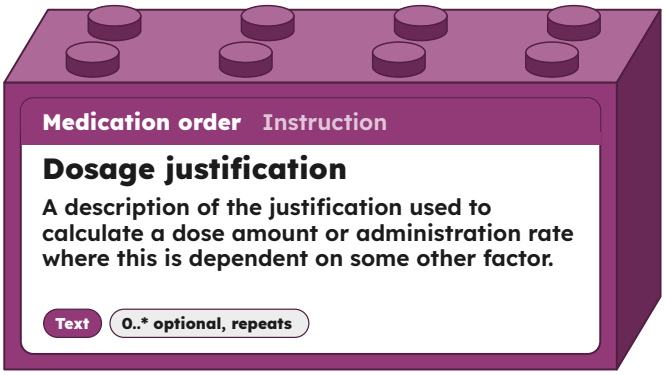




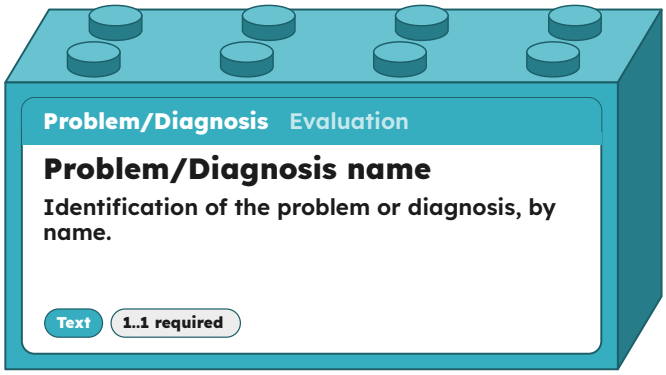


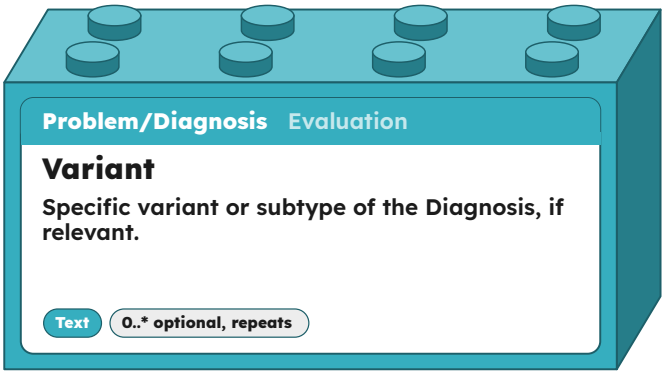


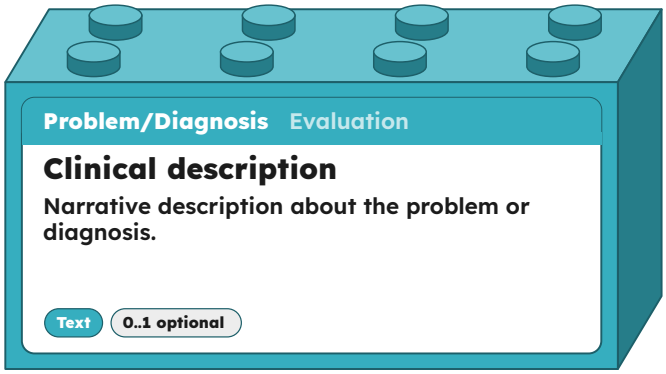


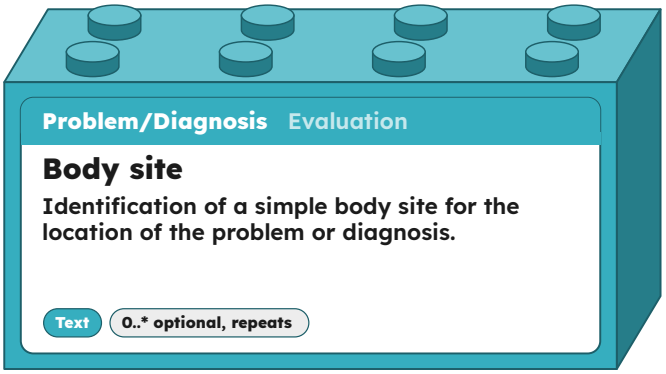


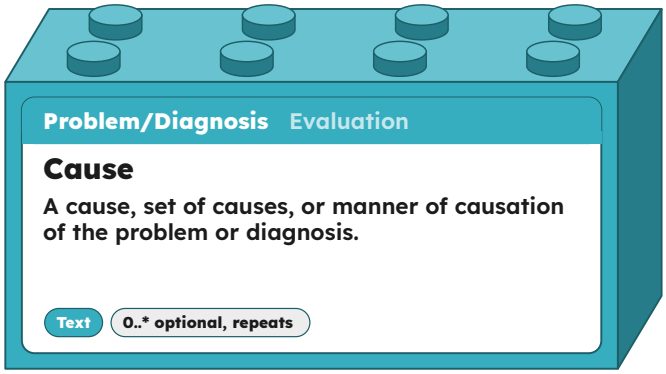
Medication order Instruction 8 of 44 elements

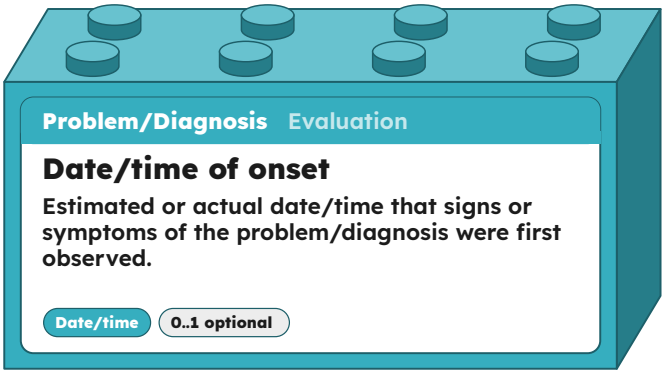


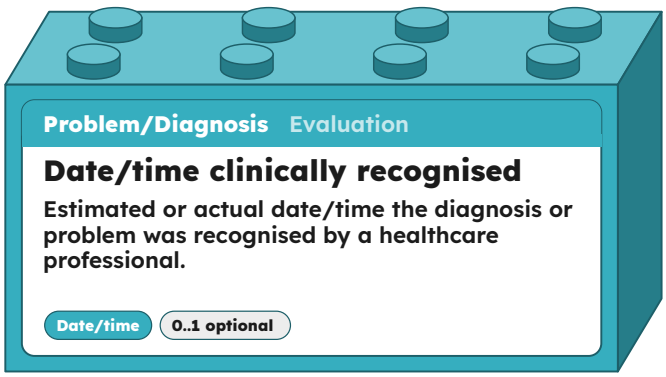


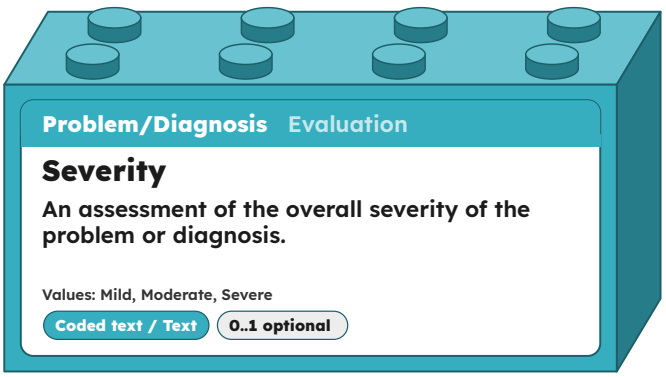




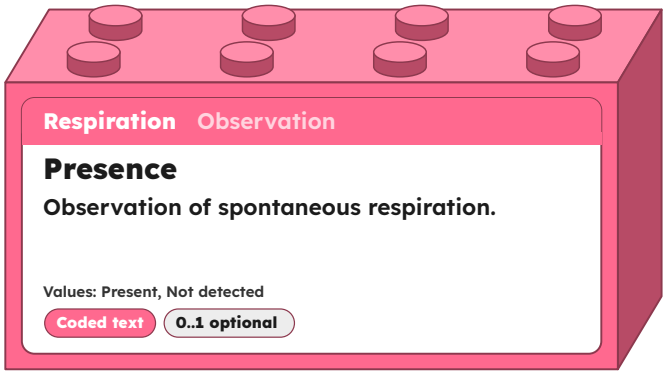


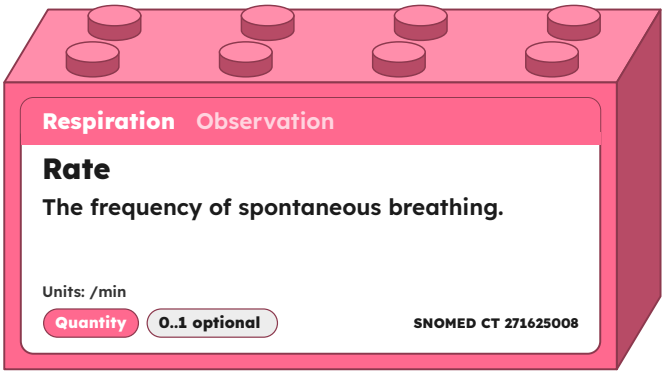


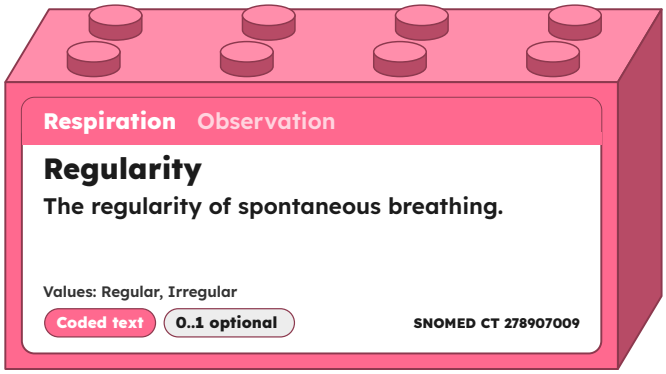


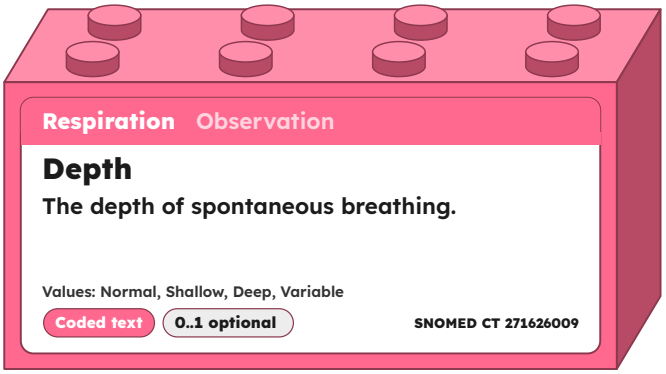


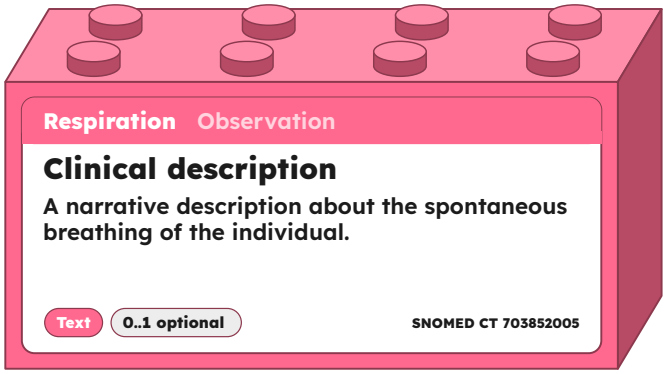
Problem/Diagnosis Evaluation 8 of 17 elements

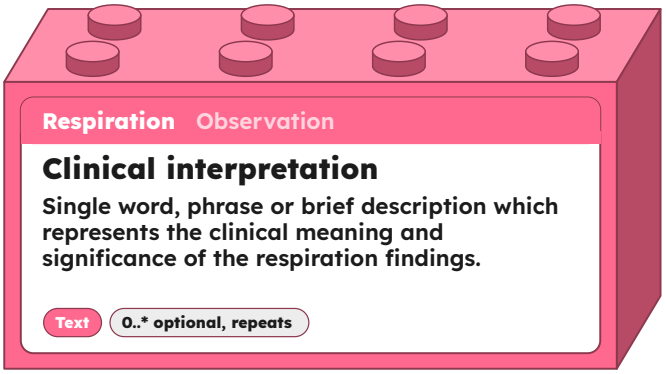


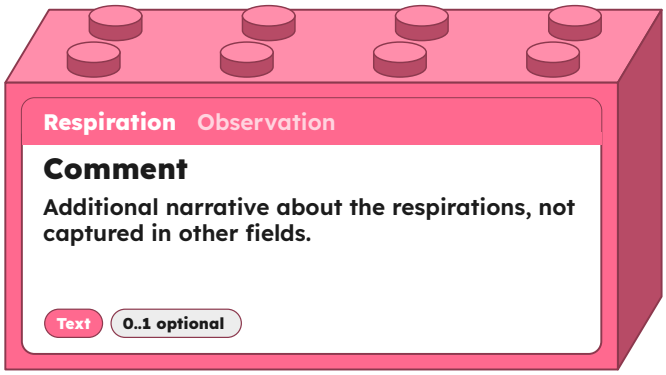


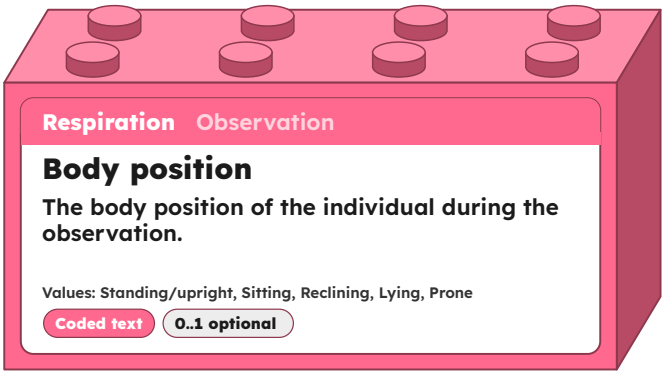




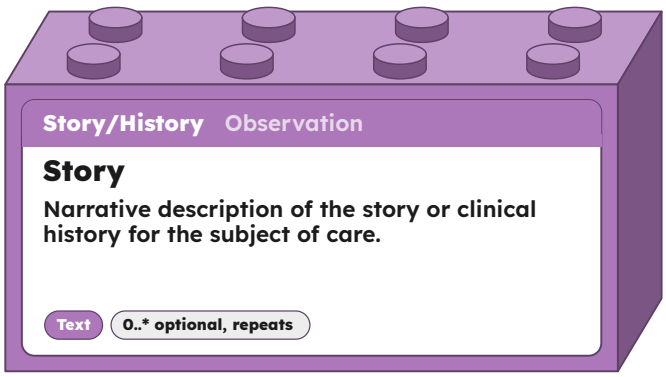


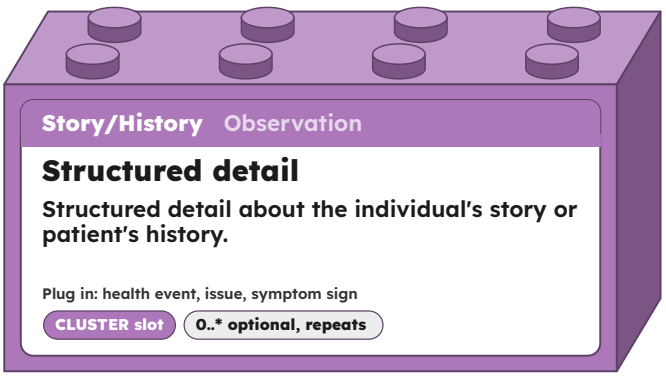






Respiration Observation 8 of 11 elements





Story/History Observation 2 of 2 elements

Template override stickers

Yellow = changed in the template. Templates can only narrow what the archetype allows.

Cardinality

Stick over the cardinality chip of a building block. Archetype cardinality goes on its plate.

1..1 required	**1..1 required**	**1..1 required**	**1..1 required**	**1..1 required**	**1..1 required**
____ .. ____	____ .. ____	____ .. ____	____ .. ____	____ .. ____	____ .. ____

Rename

Stick over the field name, or over the name of an archetype plate. Blank ones: write the new name yourself.

RENAMED IN TEMPLATE **Allergy**	RENAMED IN TEMPLATE **Reaction**
RENAMED IN TEMPLATE **Directions**	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____
RENAMED IN TEMPLATE _____	RENAMED IN TEMPLATE _____